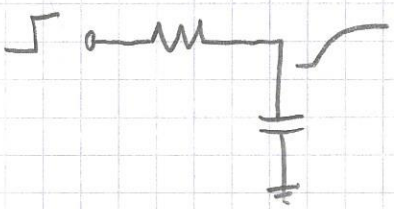
W2AEW

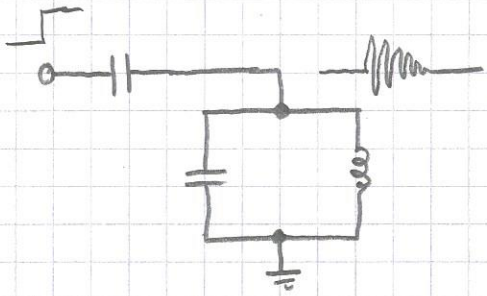
MEASURE CAPACITORS & INDUCTORS WITH YOUR OSCILLOSCOPE

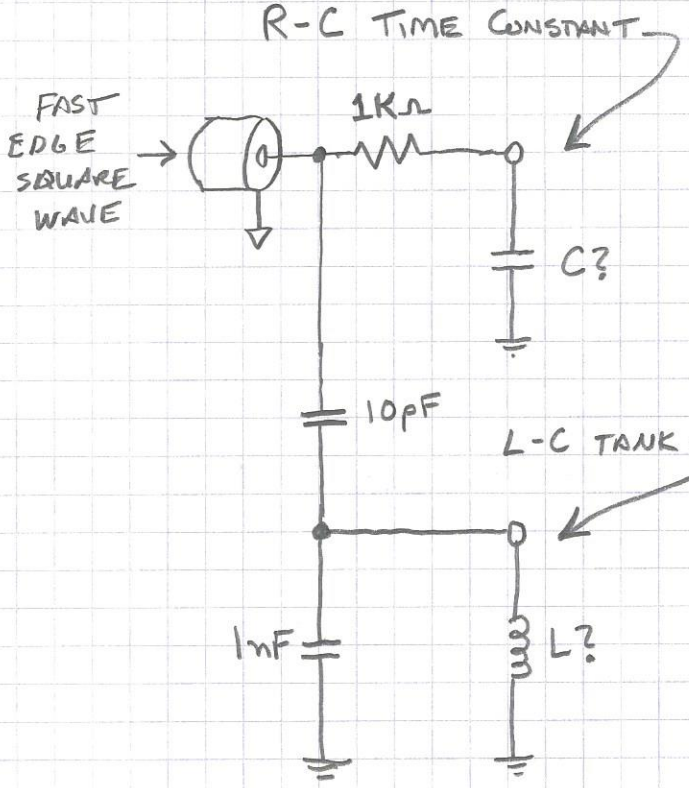
- USING THE "FAST-EDGE" PULSE GENERATOR CIRCUIT

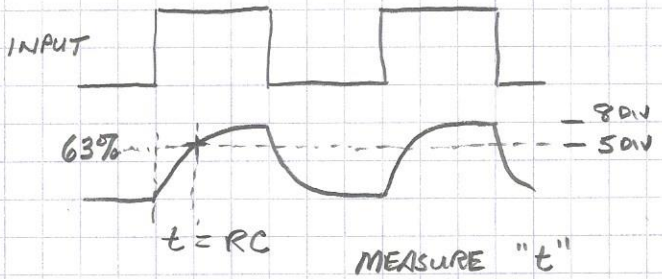
CAPACITORS



INDUCTORS

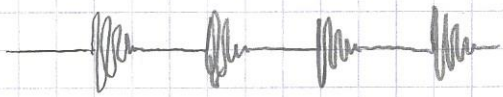






MEASURE "t"

$C = t/R$



THE TANK WILL "RING" AT ITS RESONANT FREQUENCY

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

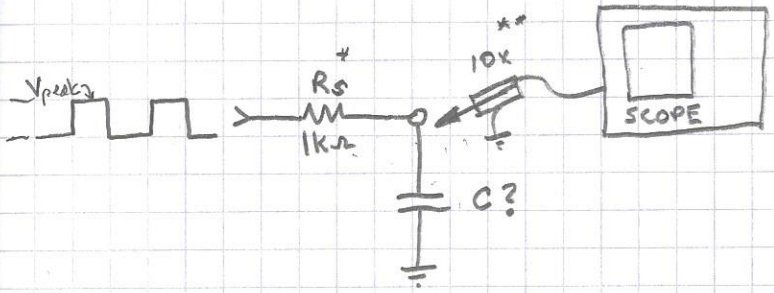
$$L = \frac{1}{(2\pi f)^2 C}$$

W2AEW

MEASURE CAPACITANCE & INDUCTANCE WITH YOUR OSCILLOSCOPE

& A SQUARE WAVE OSCILLATOR

FOR CAPACITORS...



- MEASURE R·C TIME CONSTANT

$$V = V_{peak} * \left(1 - e^{-t/RC}\right)$$

WHEN $t = RC$, THE VOLTAGE HAS RISEN 63% OF FULL VALUE

NOTES

* MAY WANT DIFFERENT VALUE

** WATCH PROBE CAPACITANCE FOR LOW C VALUES

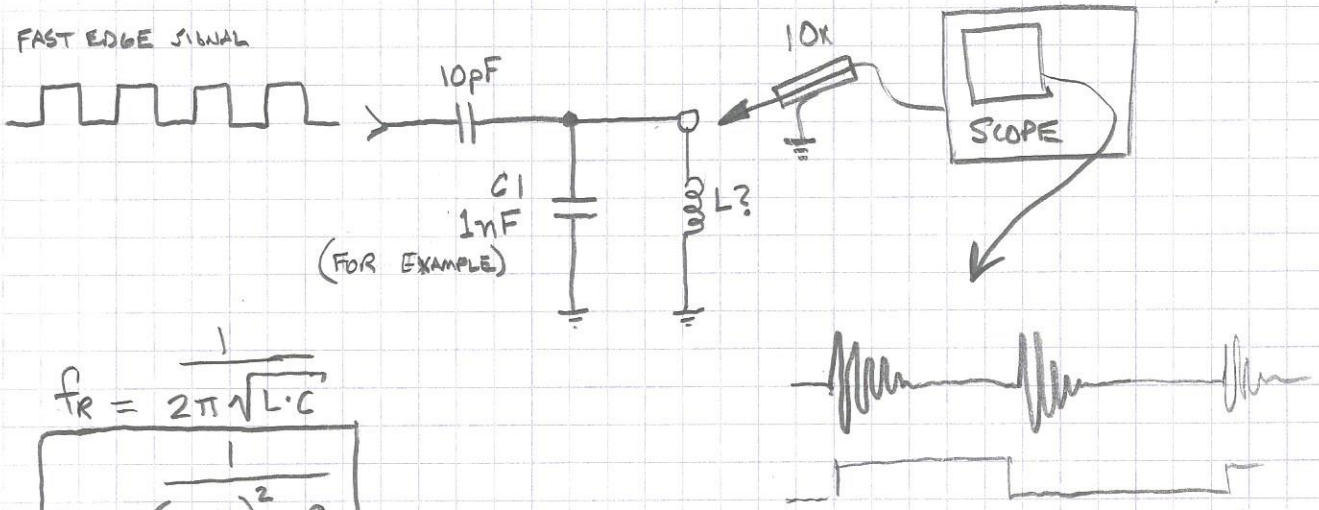
EASY WAY

- ADJUST VERTICAL TO 8 DIVISIONS
- MEASURE Δt TO RISE 5 DIVISIONS

$$C = \frac{t}{R}$$

FOR INDUCTORS... (& CAPACITORS TOO)

- START WITH A KNOWN CAPACITOR
- MAKE A PARALLEL TANK CIRCUIT
- "RING" IT WITH A FAST EDGE & MEASURE RING FREQUENCY



$$f_R = \frac{1}{2\pi\sqrt{L \cdot C}}$$

$$L = \frac{1}{(2\pi f_R)^2 \cdot C}$$